Dr. S. Balaji, Ph.D. | Mathematics

SASTRA Deemed University – Thanjavur, Tamil Nadu 613401 – India
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ORCiD Scopus Google Scholar University Profile

PROFESSIONAL SUMMARY

Distinguished mathematician with **25 years of progressive academic experience** and proven track record in research, teaching, and academic leadership. Working to leverage expertise in **algorithmic graph theory, combinatorial optimization, numerical methods, and wavelet approximation** while continuing to mentor doctoral students and contribute to institutional excellence.

Research Excellence

Publications	SCOPUS	h-index/i10-index	Citations	IF/SNIP
37 Papers	37	17/23	965	30.66/20.04

Authorship	Corresponding	First	Other	Sole
Percentage	85%	10%	5%	10 Papers

Specializations: • Algorithmic Graph Theory • Combinatorial Optimization • Numerical Methods • Wavelet Approximation

Key Publications

2025: Maheswari, G.,**Balaji, S.** "A hybrid jumping swarm optimization based local search approach for minimum weighted dominating set problem," *Evolving Systems*, 16:1-15. **IF: 2.7**

2024: **Balaji, S.,** Hariharan, G., "An efficient wavelet-based approximation method for solving nonlinear fractional-time long wave equations: An operational matrix approach," *Mathematical Methods in the Applied Sciences*, 47:1015-1033. **IF: 2.9**

2023: N. Revathi., R. Nithyakala., **Balaji, S.**, "A Bio-Inspired Technique for the Maximum Weighted Clique Problem," *National Academy Science Letters*, Article in Press. **IF: 1.2**

2021: **Balaji, S.**, Vikram, S.T., Kanagasabapathy, G., "Jumping particle swarm optimisation method for solving minimum weight vertex cover problem," *International Journal of Bio-Inspired Computation*, 18(3):143-152. **IF: 3.5**

2016: **Balaji, S.**, "A new Bernoulli wavelet operational matrix of derivative method for the solution of nonlinear singular Lane-Emden type equations arising in astrophysics," *Journal of Computational and Nonlinear Dynamics (ASME)*, 11(051013). **IF: 1.8**

Education

SASTRA Deemed University — **Thanjavur**
Ph.D. in Mathematics — *2005–2012*
Thesis: "Effective Heuristics on Graph Optimization Problems"
Selected in the pool of best theses

Madurai Kamaraj University — **Madurai**
M.Phil. Mathematics, First Class — *2003–2004*

The MDT Hindu College **Tirunelveli**
M.Sc. Mathematics, Distinction *1999–2001*
College Topper Award

Sri Paramakalyani College **Alwarkuruchi**
B.Sc. Mathematics, First Class *1996–1999*

Academic Experience

SASTRA Deemed University **Thanjavur**
Lecturer - Professor *Sept 2004–Present*
21 Years of Continuous Service

- Teaching UG (B.Tech/B.Sc) and PG (M.Tech/M.Sc) level mathematics courses
- **Doctoral Supervision:** 2 Ph.D. awarded, 2 currently supervising
- Research focus: Graph optimization, combinatorial algorithms, numerical methods
- Active contributor to NAAC & NBA accreditation processes
- Board member of syllabus revision committee
- Designed AI based syllabus for various programmes
- Course coordinator for online degree programs (BCA, M.Com)

Sri Paramakalyani College **Alwarkuruchi**
Lecturer *June 2003–Aug 2004*

- Managed B.Sc courses for regular and evening batches
- Provided additional academic support for struggling students

The MDT Hindu College **Tirunelveli**
Lecturer *Aug 2001–Apr 2003*

- Taught B.Sc and M.Sc level mathematics courses

Doctoral Supervision & Mentoring

Ph.D. Graduated: Two scholars graduated

- **Dr. S.T. Vikram** (2024) – "Strong Chromatic index of graphs derived from Path, Star, and Cycle graphs"
- *Current Position:* Assistant Professor (SG), Rajalakshmi College of Engineering, Chennai.
- **Dr. N. Revathi** (2025) – "A Computational Study On Jumping Particle Swarm-Based Heuristics For Selected Graph Optimizations Problems"
- *Current Position:* Assistant Professor, The Marudhu Pandiyar College, Thanjavur.

Ph.D. In Progress: Two in Progress and one going to submit thesis

- **G. Maheswari** – Algorithmic Graph Theory
- **V. Devi** – Algorithmic Graph Theory

M.Sc. Projects: Supervised **20 M.Sc. Data Science projects** (2020-2025) and **25 Integrated M.Sc. Mathematics projects** (2015-2025) in areas including:

- AI-driven applications (Drug-Target Interactions, Traffic Accident Prevention)
- Machine Learning (Spam Detection, Fake News Detection, Alzheimer's Prediction, Weather Forecast)
- Agricultural Data Science (Agroclimatic Clustering for Crop Yield)
- Graph Optimization, Combinatorial Optimization, Numerical Methods, Wavelet Approximations

Research Funding

Principal Investigator: **Project:** "Vertex Cover-based Link Monitoring and Backbone Formation in Wireless Ad hoc Networks"
Funding Agency: SASTRA-TRR Research Fund
Duration: Feb 2024 – Feb 2025
Amount: ₹ 1,59,000
Status: Completed successfully with three SCIE Journal publications

Awards & Recognition

2023: **Best Paper Award** – IEEE 9th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore

2018: **Certificate of Reviewing** – Journal of Computational and Applied Mathematics, Elsevier

2012: **Cash Prize Award** – ₹ 1,00,000 for timely Ph.D. completion, SASTRA University

2001: **Merit Scholarship** – College topper with distinction in M.Sc., MDT Hindu College

Academic Leadership & Service

Editorial & Review Service

- Peer Reviewer for Elsevier, Springer, Taylor & Francis, Hindawi
- Ph.D. External Examiner – Bharathiyar University, Coimbatore

Conference Organization

2020: **Webinar Coordinator** – "Mathematical Modelling and Its Applications" (9-lecture series with IIT, IISc, NIT, IMSc speakers)

2016: **Session Chair** – 19th Ramanujan Symposium, RIASM, University of Madras

Institutional Service

- NAAC & NBA accreditation coordinator
- Curriculum development for UG/PG programs
- SWAYAM-NPTEL video content creation
- Student admissions and counseling

Teaching Excellence

Academic Excellence

Student Feedback: Consistently maintaining feedback score $\geq$ **4.65 to 5** / 5

Student Engagements: Mentoring, Bridge Courses, & Coaching Classes

Course Development

Online: **MATOL120** – Inferential Discrete Mathematics (BCA program)

MCOMOL104 – Advanced Business Statistics (M.Com program)

Book Contributions

2023: Chapter: **"Vector Differentiation and Integration"** in *Mathematics for Engineers*, Pearson India

2023: Tamil Translation: **"Thisaiyan Nunganitham"**, Pearson India

2014: Chapter: **"Purpose of Evaluation"** in *SASTRA B.Ed Course Materials*, Pearson India

Invited Lectures & Presentations

Jan 2026: **One-Day Symposium** – "AI as Teaching Assistant - Practical Experiments in AI-Enhanced Pedagogy" SASTRA Deemed University (Second Teaching &LC Symposium)

July 2025: **One-Day Workshop** – "AI in Mathematics – Ideas to Experiment in 2025-26" SASTRA Deemed University (Synergistic Learning of Mathematics for Engineering Programme Students)

Dec 2024: **National Mathematics Day** – Bon Secours College for Women, Thanjavur
Theme: "Mathematics: The Bridge to Innovation and Progress"

Jan 2024: **5-Day Workshop** – "Transforming Science through Data-Driven Models" SASTRA Deemed University (Scientific Machine Learning in Ocean Engineering)

July 2024: **IEEE ICEEICT 2024** – "High-Performance Digital Filters in Electronic Stethoscope" Third International Conference, Trichy

Feb 2019: **National Conference** – "Algorithmic Graph Theory" Marudupandiyar College, Thanjavur

Mar 2016: **Ramanujan Institute (RIASM)** – "Bernoulli Wavelet operational matrix of derivatives" University of Madras, Chennai

Professional Development

2025: **AI in Industrial & Management Engineering** – NPTEL/IIT Kanpur | **Grade: Elite + Silver**

2025: **NPTEL-IIT Kanpur FDP** – AI in Industrial & Management Engineering

2024: **ChatGPT & AI Hacks with MS Office** – Microsoft Certified Training

2020: **LaTeX for Scientists and Engineers** – IIT Bombay | **Grade: A+**

2021: **AICTE-ATAL FDP** – Financial Mathematics & Machine Learning

2021: **AICTE-ATAL FDP** – Big Data and Analytics

Technical Skills

Programming: LaTeX, MATLAB, C, C++

Research Tools: Mathematical modeling, Algorithmic design, Data analysis

Emerging Tech: AI/ML applications, Data Science integration

VISION

I aim to: **(1)** Expand research impact through collaborative networks and external funding, **(2)** Increase doctoral supervision capacity and research output, **(3)** Integrate AI and data science with traditional mathematics curricula, **(4)** Bridge academic research with industry applications, and **(5)** Develop international research partnerships in algorithmic mathematics.

LIST OF PUBLICATIONS

Dr. **S. Balaji**

2025

- G. Maheswari, **S. Balaji**
  A Jumping swarm optimization-based hybrid approach for solving the minimum dominating set problem.
  2025 IEEE 2nd International Conference for Women in Computing (InCoWoCo), India, pp. 1-10, doi: 10.1109/InCoWoCo68239.2025.11407240.
- G. Maheswari, **S. Balaji**
  A hybrid jumping swarm optimization based local search approach for minimum weighted dominating set problem.
  Evolving Systems, Vol. 16, pp. 1–15.
- G. Maheswari, **S. Balaji**
  A Swarm Intelligence Technique for Link Monitoring Problems in Wireless Ad Hoc Networks.
  National Academy Science Letters, Vol. 48, pp. 613–618.

2024

- G. Maheswari, **S. Balaji**, N. Revathi, R. Nithyakala
  Jumping particle swarm-based local search approach for the maximum weighted independent set problem.
  Discrete Mathematics, Algorithms and Applications, Article in Press.
- **S. Balaji**, M. Subbarao, S. Kiruthik, V. J. M. Jerard, P. Ganesan, V. Elamaran
  High-Performance Digital Filters in an Electronic Stethoscope: An FPGA Based Approach.
  2024 Third International Conference on Electrical, Electronics, Information and Communication Technologies (ICEEICT), IEEE, pp. 1–7.
- **S. Balaji**, G. Hariharan
  An efficient wavelet-based approximation method for solving nonlinear fractional-time long wave equations: An operational matrix approach.
  Mathematical Methods in the Applied Sciences, Vol. 47, pp. 1015–1033.
- V. S. Thiru, **S. Balaji**
  Strong chromatic indices of certain binary operations on graphs.
  Discrete Mathematics, Algorithms and Applications, Vol. 16(06), 2350073.

2023

- S. T. Vikram, **S. Balaji**
  An Analysis of the Factors Influencing the Strong Chromatic Index of Graphs Derived by Inflating a Few Common Classes of Graphs.
  Symmetry, Vol. 15, 1301.
- N. Revathi, R. Nithyakala, **S. Balaji**
  A Bio-Inspired Technique for the Maximum Weighted Clique Problem.
  National Academy of Science Letters, Vol. 46(3), pp. 223–227.
- S. Fairooz, **S. Balaji**, R. Ramya, P. Thanapal, V. Elamaran

A Case Study using Simple Moving Average Filters to accomplish ECG denoising on an FPGA.
9th International Conference on Advanced Computing and Communication Systems (ICACCS 2023), IEEE, pp. 244–248.

- T. R. Reddy, **S. Balaji**, R. Ramya, P. Thanapal, V. Elamaran
  Analyzing Data Compression Techniques for Biomedical Signals and Images Using Downsampling and Upsampling.
  9th International Conference on Advanced Computing and Communication Systems (ICACCS 2023), IEEE, pp. 71–76. (Best Paper Award)

2022

- **S. Balaji**, S. T. Vikram
  Strong chromatic index of generalized Jahangir graphs and generalized Helm graphs.
  Discrete Mathematics, Algorithms and Applications, Vol. 14, 2250045.

2021

- **S. Balaji**, S. T. Vikram
  Strong Chromatic index of generalized polygon snake graphs.
  National Academy of Science Letters, Vol. 44(2), pp. 151–154.
- **S. Balaji**, S. T. Vikram, G. Kanagasabapathy
  Jumping particle swarm optimisation method for solving minimum weight vertex cover problem.
  International Journal of Bio-Inspired Computation, Vol. 18(3), pp. 143–152.

2020

- **S. Balaji**, K. Naganthran, A. Zeeshan, Md. Basir, N. Shehzad, R. Nazar, R. Choudary
  Concentration Flux Dependent on Radiative MHD Casson Flow with Arrhenius Activation Energy: Homotopy Analysis Method (HAM) with an Evolutionary Algorithm.
  International Journal of Heat and Technology, Vol. 38(4), pp. 785–793.
- **S. Balaji**, N. Rajagopal, R. Seethalakshmi, V. S. Balaji
  A new numerical method for fractional order Volterra integro-differential equations.
  Ain Shams Engineering Journal, Vol. 11, pp. 171–177.

2019

- **S. Balaji**, G. Hariharan
  An efficient operational matrix method for the numerical solutions of the fractional Bagley–Torvik equation using wavelets.
  Journal of Mathematical Chemistry, Vol. 57, pp. 1885–1901.

2018

- **S. Balaji**, V. Venkataraman, D. R. V. S. R. K. Sastry
  Thermal and solutal dispersion effects on heat and mass transfer in a viscous MHD flow over a vertical plate.

International Journal of Pure and Applied Mathematics, Special Issue.

- **S. Balaji**, G. Hariharan
  A novel wavelet approximation method for the solution of nonlinear differential equations with variable coefficients arising in astrophysics.
  Astrophysics and Space Science, Vol. 363(16).

2017

- **S. Balaji**, V. Venkataraman, D. R. V. S. R. K. Sastry
  Hydrodynamic marangoni convection in a nanofluid flow with soret and dufour effects.
  International Journal of Civil Engineering and Technology, Vol. 8, pp. 164–171.
- **S. Balaji**, V. Venkataraman, D. R. V. S. R. K. Sastry
  Solution of system of nonlinear equations using integrated RADM and ADM.
  International Journal of Pure and Applied Mathematics, Vol. 117, pp. 367–373.

2016

- N. Revathi, **S. Balaji**
  A new approach for solving set covering problem using Jumping particle swarm optimization method.
  Natural Computing (Springer), Vol. 15, pp. 503–517.
- **S. Balaji**
  A new Bernoulli wavelet operational matrix of derivative method for the solution of nonlinear singular Lane–Emden type equations arising in astrophysics.
  Journal of Computational and Nonlinear Dynamics (ASME), Vol. 11(051013).
- **S. Balaji**, V. Venkataraman, D. R. V. S. R. K. Sastry
  Effect of radiation on mhd mixed convection flow past a semi infinite vertical plate.
  ARPN Journal of Engineering and Applied Sciences, Vol. 11, pp. 14033–14038.

2015

- **S. Balaji**
  Legendre wavelet method for solution of nonlinear Riccati differential equation.
  Journal of Egyptian Mathematical Society, Vol. 23, pp. 263–270.

2014

- **S. Balaji**
  A new approach for the solution of duffing equation involving both integral and non-integral forcing terms.
  Ain Shams Engineering Journal, Vol. 5, pp. 985–990.
- **S. Balaji**
  Solution of nonlinear Riccati differential equation using Chebyshev wavelets.
  WSEAS Transaction on Mathematics, Vol. 13, pp. 441–451.
- **S. Balaji**
  Legendre wavelet method for solution of nonlinear Riccati differential equation of fractional order.
  International Journal of Mathematics and Mathematical Sciences, Vol. 10, 304745.

2013

- **S. Balaji**, Y. B. Venkatakrishnan, K. Kannan
  Total dominating set based algorithm for minimum connected dominating set in ad hoc wireless networks.
  WSEAS Transaction on Mathematics, Vol. 12, pp. 1164–1172.
- **S. Balaji**
  A local search based greedy heuristic for the minimum vertex cover problem.
  Far East Journal of Mathematical Sciences, Vol. 79, pp. 257–266.
- **S. Balaji**
  A new effective local search heuristic for the maximum clique problem.
  International Journal of Mathematical Sciences, Vol. 7, pp. 32–38.

2012

- **S. Balaji**, N. Revathi
  An efficient approach for the optimization version of the maximum weight clique problem.
  WSEAS Transaction on Mathematics, Vol. 11, pp. 773–783.
- **S. Balaji**, N. Revathi
  An efficient heuristic for the minimum connected dominating set on ad hoc wireless network.
  International Journal of Computer and Communication Engineering, Vol. 6, pp. 278–284.

2010

- **S. Balaji**, V. Swaminathan, K. Kannan
  A Simple Algorithm for Maximum Clique and Matching Protein Structures.
  International Journal of Combinatorial Optimization Problems and Informatics, Vol. 1, pp. 2–11.
- **S. Balaji**, V. Swaminathan, K. Kannan
  Optimization of Unweighted Minimum Vertex Cover.
  Proceedings of World Academy of Science Engineering and Technology, Vol. 67, pp. 508–513.
- **S. Balaji**, V. Swaminathan, K. Kannan
  An Effective Algorithm for minimum weighted vertex cover problem.
  International Journal of Computational and Mathematical Sciences, Vol. 4, pp. 34–38.
- **S. Balaji**, V. Swaminathan, K. Kannan
  A Simple Algorithm to Optimize Maximum Independent Set.
  Advanced Modeling and Optimization, Vol. 12, pp. 107–118.

2009

- **S. Balaji**, V. Swaminathan, K. Kannan
  Approximating Maximum Weighted Independent Set Using Vertex Support.
  International Journal of Computational and Mathematical Sciences, Vol. 3, pp. 412–417.

Summary

Total Publications: 37
SCI/SCIE Journals: 18 — SCOPUS/WOS Journals: 37
Webpage Links:
ORCID | Scopus | Google Scholar | University Profile

My Progress in AI Usage

A Continuing Journey of Learning, Creating & Innovating

Apps & Software Tools

Developed **two application/software tools** (mobile/web) using Claude-AI:

- **SASTRA Mentor Connect APP** — an easy-to-use student mentoring platform.
- **Resume Analyzer APP** — compares a student's résumé against a Job Description (JD) and highlights missing keywords/skills.

FDP & Course Certificates

- **Feb 2024:** Completed AI workshop on *ChatGPT & AI Hacks with MS Office*, conducted by Mr. Jatan Shah — Microsoft Certified Trainer, Co-founder & CEO, Skill Nation.
- **Aug–Oct 2025:** Completed **NPTEL-AICTE Faculty Development Programme** for the course *Artificial Intelligence in Industrial and Management Engineering*, conducted by **IIT Kanpur**.
- **Aug–Oct 2025:** Secured **Elite + Silver Grade** in the proctored exam of *Artificial Intelligence in Industrial and Management Engineering*, IIT Kanpur.

Advanced AI Curriculum

- **July 2025:** Framed an AI-enhanced curriculum for the *B.Tech (ECE) V Semester* course — **Probability and Random Processes with AI Integration**.
- **Dec 2025:** Framed an AI-enhanced curriculum for *B.Sc., M.Sc. Mathematics and Data Science* courses under SASTRA University — Zoho initiative of **Building Mathematical Foundations for Modern AI**.

Invited Lectures

- **Jan 2025:** Presented a lecture and hands-on session titled *"Transforming Science through Data-Driven Models"* at the **Five-Day Workshop on Scientific Machine Learning in Ocean Engineering**, SASTRA University.
- **July 2025:** Presented a lecture on *"AI in Mathematics — Ideas to Experiment in 2025–26"* at the **Workshop on Synergistic Learning of Mathematics for Engineering Programme Students**, SASTRA University.
- **Jan 2026:** Presented a lecture titled *"AI as Teaching Assistant — Practical Experiments in AI-Enhanced Pedagogy"* at the **Second TLC Symposium**, SASTRA University.

Research Papers

Three papers with the following titles are **under review** in **SCIE Journals**:

- **An AI-Driven Interactive Interface for Interview Excellence**
- **Rainfall Forecasting Using Machine Learning and Time Series Models**
- **Enhanced Rainfall Prediction using Machine Learning and Actionable Insights**

Academic Activities

- Use of AI tools such as **Google Colab**, **Desmos**, etc., for classroom teaching.
- Use of AI tools such as **Keptune**, **ChatGPT**, **Claude**, etc., for School Result Analyses.

. . . and the journey is *still counting* . . .

(Signature)
Dr. S. Balaji